Big Data HomeWork 1

Lekhana Sandra

Step-by-Step Guide for Setting Up AWS Academy Learner Lab (With Screen snips attached below)

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1. Open AWS Academy Learner Lab from Modules

- Navigate to your **AWS Academy course dashboard**.
- Click on **"Modules"** in the navigation panel.
- Select **"AWS Academy Learner Lab"** and open it.

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2. Start AWS Academy Learner Lab from Modules

- Click **"Start Lab"** to initiate the environment.
- Wait for the lab to **provision resources** and become active.

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3. Choose Two Instances and Assign Names

- Open the **AWS Management Console**.
- Navigate to **EC2 Dashboard**.
- Click **"Launch Instance"** and select **two instances**.
- Assign meaningful names like **"Instance1"** and **"Instance2"** under the **Tags** section.

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4. Generate a Key Pair, Download the .PEM File, Copy the Directory Path, and Open in Terminal

- In the **Key Pair** section, click **"Create new key pair"**.
- Choose **RSA** as the key type and enter a name (e.g., `MyKeyPair`).
- Click **"Create"** and **download the .PEM file**.

Open **Terminal** and navigate to the directory where the `.PEM` file is stored:
cd /path/to/your/keypair

- Set the correct file permissions:
  chmod 400 MyKeyPair.pem

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5. Launch the Instance Successfully

- Click **"Launch Instance"** and wait for it to **initialize**.
- Verify that the instances are in the **"Running"** state in the EC2 Dashboard.

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6. Click on Connect

- In the **EC2 Dashboard**, select an instance.
- Click the **"Connect"** button at the top.
- Choose **"SSH Client"** and follow the connection instructions.

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7. Copy the Provided Example and Execute in Terminal

Copy the given SSH connection command, for example:
ssh -i "MyKeyPair.pem" ubuntu@<Public_IP_of_Instance>

- Paste it into **Terminal** and press **Enter**.

aws | Search [Option+S] | United States (N. Virginia) | voclabs/user3841815=Lekhana_Sandra @ 9687-1529-6341

- Dashboard
- EC2 Global View
- Events
- **Instances**
  - Instances
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  - Savings Plans
  - Reserved Instances
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  - AMIs
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- **Elastic Block Store**
  - Volumes
  - Snapshots
  - Lifecycle Manager
- **Network & Security**
  - Security Groups
  - Elastic IPs

Resources

EC2 Global View

You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:

Instances (running)	9	Auto Scaling Groups	0	Capacity Reservations	0
Dedicated Hosts	0	Elastic IPs	0	Instances	9
Key pairs	5	Load balancers	0	Placement groups	0
Security groups	12	Snapshots	0	Volumes	9

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instance

Migrate a server

Note: Your instances will launch in the United States (N. Virginia) Region

Instance alarms

View in CloudWatch

0 in alarm | 0 OK | 0 insufficient d ata

Service health

AWS Health Dashboard

Region
United States (N. Virginia)

Status
This service is operating normally.

Zones

Zone name	Zone ID
us-east-1a	use1-az6
us-east-1b	use1-az1

Account attributes

Default VPC
vpc-000ca9018818b412a

Settings
- Data protection and security
- Allowed AMIs
- Zones
- EC2 Serial Console
- Default credit specification
- EC2 console preferences

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EC2 > Instances > Launch an instance

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Browse more AMIs
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type — Free tier eligible
ami-04b4f1a9cf54c11d0 (64-bit (x86)) / ami-0a7a4e87939439934 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Description
Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture	AMI ID	Username
64-bit (x86)	ami-04b4f1a9cf54c11d0	ubuntu Verified provider

Summary

Number of instances Info
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-04b4f1a9cf54c11d0

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and

Cancel | Launch instance | Preview code

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EC2 > Instances > Launch an instance

On-Demand Linux base pricing: 0.0116 USD per Hour

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

MYkey7 | Create new key pair

Network settings Info

Edit

Network Info

vpc-000ca9018818b412a

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- (•) Create security group
- ( ) Select existing security group

We'll create a new security group called '**launch-wizard-9**' with the following rules:

[x] Allow SSH traffic from

Summary

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Canonical, Ubuntu, 24.04, amd6...read more
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Cancel | Launch instance | Preview code

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EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-02cf8743a7a6b9c11)

▸ Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

1 2 3 4 5 6

Create billing and free tier usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the

Create Load Balancer

Create a application, network gateway or classic Elastic Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

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EC2 > Instances

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Elastic IPs

Instances (1/9) Info

Last updated less than a minute ago | Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | All states | 1

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv
datanode1	i-022003817a2715112	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-44-20
Unselect instance: datanode1	i-036fcaa2f1ba6f3e3	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-52-87
datanode3	i-0c7e5dd311d8eb19a	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-18-21
	i-05141ec3301601d77	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-44-20
	i-009c2a3be8bb9586d	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-18-20
	i-09f23adae5fcbc795	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-54-85
	i-044d5574bf985e9e8	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-54-21
	i-0b3cea6b30f94a8e5	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1c	ec2-34-23

i-022003817a2715112 (datanode1)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary Info

Instance ID	Public IPv4 address	Private IPv4 addresses
i-022003817a2715112	44.204.34.52 | open address	172.31.93.63
IPv6 address	**Instance state**	**Public IPv4 DNS**

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EC2 > Instances > i-022003817a2715112 > Connect to instance

Connect to instance Info

Connect to your instance i-022003817a2715112 (datanode1) using any of these options

EC2 Instance Connect | Session Manager | SSH client | EC2 serial console

Instance ID
i-022003817a2715112 (datanode1)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is lab1.pem
3. Run this command, if necessary, to ensure your key is not publicly viewable.
   chmod 400 "lab1.pem"
4. Connect to your instance using its Public DNS:
   ec2-44-204-34-52.compute-1.amazonaws.com

Example:
ssh -i "lab1.pem" ec2-user@ec2-44-204-34-52.compute-1.amazonaws.com

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

```
The authenticity of host 'ec2-3-95-229-238.compute-1.amazonaws.com (3.95.229.238)' can't be established.
ED25519 key fingerprint is SHA256:uPCvf/XdXVIusqikxMf4ZbNClNow0fdJSJRbOn7KdJA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-95-229-238.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Feb 21 21:34:20 UTC 2025

  System load:  0.0               Processes:             104
  Usage of /:   24.9% of 6.71GB   Users logged in:       0
  Memory usage: 20%               IPv4 address for enX0: 172.31.88.206
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```